ML Assisted Accelerometer Error Reduction

Ryan Waterman – Documentation Lead

Ngongha Fon – Communication Lead

Team Alpha

https://github.com/DSE6311

Team Alpha is interested in identifying whether high-accuracy accelerometers can be calibrated against thermal drift utilizing a machine learning model. Accelerometers embedded in Inertial Measurement Units (IMU's) are subject to erroneous output due to internal stresses from thermal expansion and contraction. IMUs, particularly in Integrated Circuit (IC) form, are ubiquitous in embedded systems from drones to phones, serving the critical function of measuring 6 degree-of-freedom motion. In highly precise applications, error in the accelerometer measurement as a result of thermal variance can lead to system level performance issues.

The primary stakeholders for this model are robotics companies, industrial and structural monitoring companies, and, more broadly, the aerospace industry as a whole. In each of these applications, precision and accuracy of the accelerometer are critical. It is likely that this subject has been explored by these stakeholders using traditional algorithms. Where machine learning approach differs is in flexibility and generalization. Instead of a unique data set being captured for each sensor and tuned accordingly, a set of representative units can be used to derive a generalized model. Team Alphas hypothesis is that the relationship between thermal energy and axial stress is a function of the substrate surface area/geometry, which is predictable and consistent in this application. Based on this hypothesis, Team Alpha predicts, given enough clean data, a model can sufficiently predict the offset vector required to cancel error due to thermal drift.

During the preliminary research phase, a dataset entitled "Temperature characterization of an high sensitivity tri-axial accelerometer through multiple thermometers" was identified from Iafolla et. al. This data can be found, here. This data contains accelerometer output data in X, Y, and Z cartesian axes, along with temperature data from a combination of 11 points on the accelerometer and in the ambient environment. Each of the three CSV files in the dataset contain ~1,000,000 samples, providing sufficient data for model training. There are two approaches for model training: developing three models to predict the error on each of the X, Y, and Z axes, or developing one model to predict the 3-dimensional error vector. The second approach is preferable to reduce training time and simplify model deployment. In this case, the target variable will be the vector representing the combined error in each axis. The temperature measurements will be the feature variables.

The preliminary analysis plan is to start with a generalized linear model (GLM) as a baseline. This will aid in determining the relationship between the variables and indicate whether the vector offset approach is viable. The next step would be to test the GLM performance against a custom deep neural network trained on the same data. Preprocessing would include determining the error vector representation from the X, Y, and Z axis measurement data. This is a two-step process, as the error in the measurement needs to first be determined. A simple approach would be to identify deviation from the mean as error, but this may be too simplistic. A better method would be to extract the mean axial acceleration when the temperature is within a percentage of the mean at each measurement interval. This would give a nominal measurement for which deviation can be classified as error. Either approach must be attempted and tested for efficacy. Determining the baseline performance of the accelerometer is the sole point of failure for the model. Poor judgement in this step can lead to classifying valuable, accurate acceleration data as error, reducing the model's efficacy. Determination of whether the research question was answered can be measured by an error prediction accuracy of over 95% from the finalized model. This is a metric that can be directly tested once the model has been finalized.

Python will be the primary programming language for this application. Packages such as numpy, pandas, scikit-learn, and tensorflow will be pivotal during data analysis and model training. Additionally, CUDA drivers from Nvidia are required to expedite the model training process using GPU acceleration.

Citations:

Iafolla, Lorenzo; Santoli, Francesco; Carluccio, Roberto; Chiappini, Stefano; Fiorenza, Emiliano; Lefevre, Carlo; Lucente, Marco; Pignatelli, Alessandro; Chiappini, Massimo (2023), "Temperature characterization of an high sensitivity tri-axial accelerometer through multiple thermometers", Mendeley Data, V1, doi: 10.17632/f78bmhr628.1